

Supplementary Material for “Exact Robust Local Graph-Smoother Design Under Hidden-Topology Completions”

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I. FIXED-KNOWN-GRAPH LOCAL ALGORITHMS

This track deliberately studies a different task from robust completion design: the full weighted graph is revealed and each method approximates that one graph’s smoother. We average over random-cyclic, grid, geometric, and small-world weighted graphs with 36 vertices and 16 graph-signal draws per point. Ordinary induced-neighborhood truncation, Neumann iteration, and Chebyshev iteration all obey the stated causal radius, but truncation also collects a local model and performs a local solve. At four rounds, their mean empirical MSEs are 8.63×10^{-5} , 6.69×10^{-2} , and 5.95×10^{-3} , respectively; corresponding mean byte-hop counts are 426,972 for truncation and 79,360 for each polynomial method. These are graph-signal approximation errors, not WLS or Kalman measurement-noise MSEs.

II. EXTERNAL SCALE STRESS TEST

Six PGLib-OPF v23.07 networks provide weighted topologies and three SuiteSparse matrices provide symmetric sparsity graphs. In every case the benchmark constructs the unit-grounded matrix $I + L$; it does not reproduce or claim the source dataset’s original estimator. The largest case has 129,164 vertices, and every sparse global-reference solve meets its configured residual tolerance. Figure 2 records 16-round Chebyshev relative signal error and reference-solve time. Neither quantity is converted into a theorem-matched hidden-completion certificate.

III. SAMPLING PROTOCOL AND METRICS

For each sampled rule Q_N , let $R_N(Q_N)$ denote the maximum full-input operator-norm error over its N training completions, and let $R_h(Q_N)$ denote the maximum over the complete finite- h scenario set. With R_h^* the robust-design optimum, the reported quantities are

$$u_N = \frac{R_h(Q_N) - R_N(Q_N)}{R_h(Q_N)}, \quad g_N = \frac{R_h(Q_N)}{R_h^*} - 1. \quad (1)$$

We declare underreporting when $R_h(Q_N) - R_N(Q_N) > \max\{2 \times 10^{-5}, 10^{-3}R_h(Q_N)\}$. The positive-contraction overhead uses the same ratio g_N with Q_N replaced by its baseline rule.

Each completion family cycles deterministically through random-cyclic, grid, and geometric topologies, while hidden

count cycles through $0, \dots, h$. Every nonzero edge weight is sampled log-uniformly on $[10^{-2}, 10^2]$; the random-cyclic and geometric generators request mean degree four. The breadth grid uses five independently derived seeds and $N = 32$ for each of 54 cells. The budget study uses 30 independent base seeds for each of two problems and nests sample prefixes for $N \in \{4, 8, 16, 32, 64, 128, 256\}$. Its Wilson bounds describe the 30 outcomes within a fixed problem and budget. The empirical CDFs pool grid evaluations without IID or population-inference claims: the positive-contraction rule is reused across the three hidden budgets, and five sampling seeds share each problem definition.

The positive-contraction baseline minimizes the same local-rule objective over the outer uncertainty set $0 \preceq X \preceq I$. Its implementation uses the lossless single-full-block Petersen LMI, including the scalar nonnegative multiplier, and then reevaluates the returned Q over every exact finite- h scenario. Thus its plotted value is finite-budget true risk, not the outer set’s optimization epigraph.

IV. CONIC-SOLVER POSTSOLVE AUDIT

The fixed 54-problem core grid is solved with CVXPY 1.7.5 and the Clarabel backend. 40 solves return `optimal` and 14 return `optimal_inaccurate`. We therefore do not use the conic epigraph variable as the reported certificate: each returned matrix is thresholded at relative magnitude 2×10^{-5} , then reevaluated directly over every exact finite scenario. Across the grid, the largest difference between the solver epigraph and the recomputed squared risk is 4.63×10^{-8} , the largest radius change caused by deployment thresholding is 6.31×10^{-9} , and the smallest deployed certificate eigenvalue is -8.87×10^{-16} . The archived record contains each status, active scenario, raw and deployed matrix, and all postsolve diagnostics.

This primal postsolve establishes the achieved radius and numerical feasibility of each returned rule; by itself it does not turn an `optimal_inaccurate` status into a numerical optimality proof. We therefore rerun all 14 flagged cells independently with SCS 3.3.1 at `eps=1e-8`. All return `optimal`. After exhaustive scenario reevaluation, the maximum relative radius disagreement with Clarabel is 2.40×10^{-8} (or 1.68×10^{-8} after deployment thresholding); the largest SCS epigraph-recomputed squared-risk gap is 6.45×10^{-10} , and the largest local-rule residual is 3.05×10^{-11} . This is numerical cross-solver corroboration, not a rigorous dual lower bound. Because Q can vary along flat optimizer directions, we

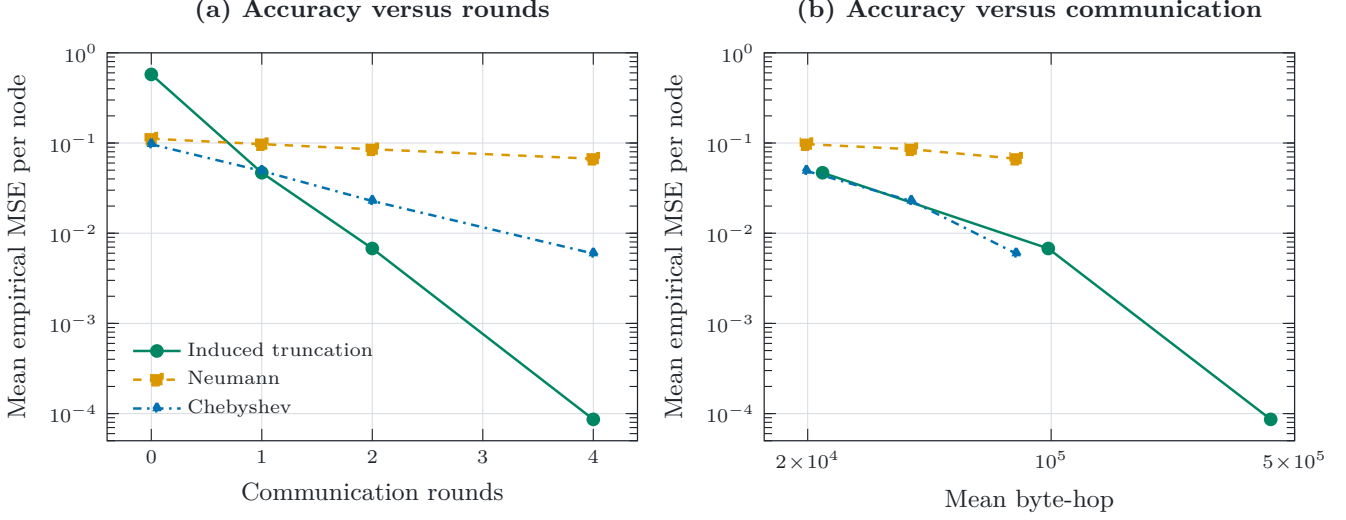


Fig. 1. Fixed-known-graph accuracy and communication tradeoff. Induced truncation attains the lowest error in this small test but collects more model traffic; Chebyshev improves faster than Neumann at the same iterative byte-hop count.

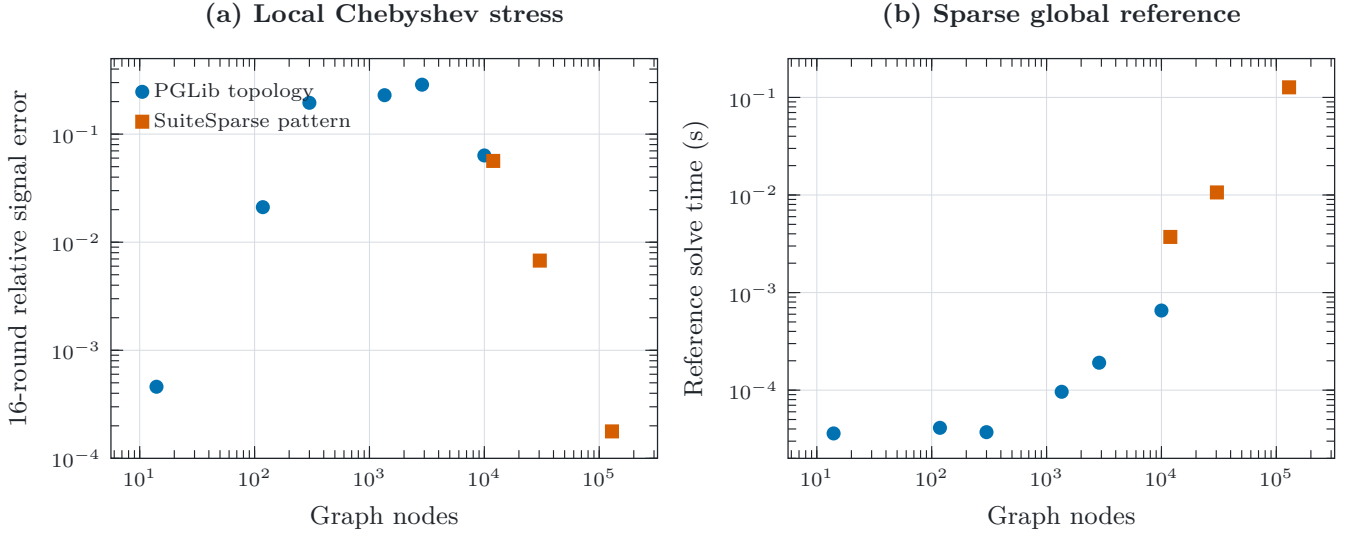


Fig. 2. External graph-smoother stress tests. PGLib cases contribute topology and positive branch weights; SuiteSparse cases contribute sparsity graphs. Panel (a) reports 16-round Chebyshev relative signal error and panel (b) the sparse global-reference solve time.

compare objective radii and make no uniqueness claim for the returned matrix. Independently of this audit, all nine one-hop state problems, the representative $q = 4, h = 1$ rule, and both exact references in the sampling-budget study already have primary status *optimal*; every zero-hop constant-preserving problem fixes $Q = I$.

V. HARDWARE-SPECIFIC ACCELERATOR TIMING

This experiment is separated from the main theorem-matched results because it measures one implementation on one platform; it is neither a complexity claim nor evidence for the completion theorem. Three independent process runs compare 40 rounds of a fixed local Richardson smoother on two-dimensional grids. OpenMP and OpenBLAS are restricted

to one CPU thread, and the accelerator is an NVIDIA RTX 5080. Median resident-kernel speedups (CPU time divided by GPU time) are 0.273, 1.468, 3.235, and 69.38 at 10,000, 50,176, 102,400, and 1,000,000 states, respectively. Including vector host-device transfers gives 0.259, 1.400, 2.922, and 50.35. The observed crossover therefore lies between 10,000 and 50,176 states on this system. The archived experiment record contains the three process-level repetitions, package and platform versions, normalized configuration, and source hash used to generate the figure.

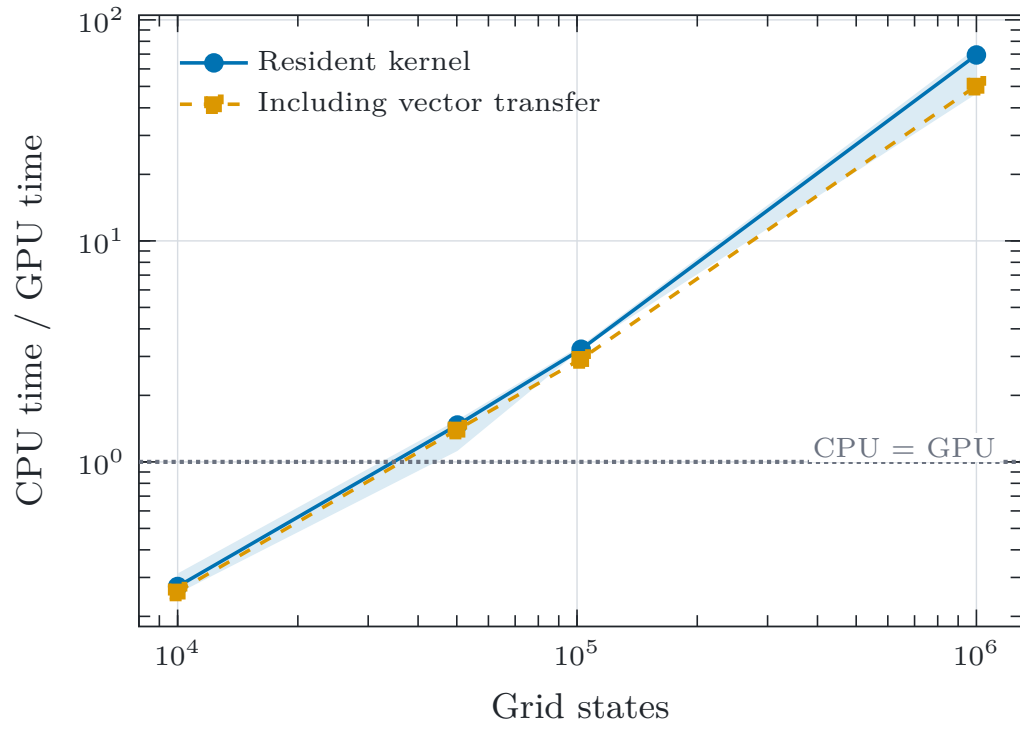


Fig. 3. CPU/GPU crossover for a 40-round unit-grounded grid smoother. Curves show medians over three process runs, and the blue band spans the observed resident-kernel range.